

NETWORKWALKS ACADEMY

Cybersecurity Internship Programme

WEEK THREE PROJECT REPORT

Password Cracking and PDF Password Recovery

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Project Area	Cybersecurity / Ethical Hacking
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Module	Tool(s) Used	Target File	Recovered Password
Module One	John the Ripper (Jumbo) + Johnny GUI	Locked PDF 3	1qaz2wsx
Module Two	Networkwalks Hash Calculator + Password Cracker	Locked PDF 2	password1

This report documents the practical laboratory exercises completed during Week Three of the Networkwalks Academy cybersecurity internship, covering local and web-based password-recovery workflows performed strictly on supplied training files within an authorized laboratory environment.

TABLE OF CONTENTS

1. Executive Summary
 2. Introduction
 3. Project Objectives
 4. Technical Background
 5. Module One — John the Ripper & Johnny Password Cracker (Locked PDF 3)
 6. Module Two — Networkwalks Password Cracker (Locked PDF 2)
 7. Comparison of Module One and Module Two
 8. Observations and Findings
 9. Skills Acquired
 10. Challenges Encountered
 11. Defensive Security Recommendations
 12. Ethical and Legal Considerations
 13. Overall Project Results
 14. Conclusion
 15. References
- Appendix A — Practical Evidence Index

1. EXECUTIVE SUMMARY

During Week Three of my cybersecurity internship at Networkwalks Academy, under the guidance of my tutor Waqas Karim, I completed a practical laboratory project focused on **password cracking and password recovery for encrypted PDF documents**. The purpose of the exercise was to understand how cybersecurity professionals test the strength of passwords protecting digital files within an authorized laboratory environment, and to gain hands-on exposure to the relationship between password-protected files, cryptographic hash representations, wordlists, password-cracking software, and result validation.

The project was divided into two practical modules. **Module One** made use of **John the Ripper (Jumbo)** together with the **Johnny graphical interface** on a Windows workstation. The password-protected file, referred to as *Locked PDF 3*, was submitted to a pdf2john-compatible online hash extractor to obtain a crackable hash, which was then saved as a text file and loaded into Johnny to run a controlled password-recovery attack. The attack successfully recovered the password **1qaz2wsx**, and the corresponding laboratory flag **nw{networkwalks_flag_260821_1}** was captured after the recovered password was used to unlock the file.

Module Two made use of the Networkwalks browser-based password-cracking laboratory. The second protected file, *Locked PDF 2*, was submitted to the Networkwalks Hash Calculator to obtain a PDF-compatible hash, which was then passed to the Networkwalks Password Cracker. The tool successfully identified the password **password1**, which was subsequently used to open the original protected PDF, confirming recovery and yielding the completion flag **nw{networkwalks_persistence_jtr_270521}**.

Both exercises reinforced an important cybersecurity principle: **weak and predictable passwords are highly vulnerable to offline password-recovery attacks**. The project provided practical experience with John the Ripper, the Johnny GUI, PDF hash extraction, wordlist-based recovery, web-based password auditing tools, evidence collection, and result validation — all performed exclusively on the supplied training files and within the defined scope of the Networkwalks cybersecurity laboratory.

2. INTRODUCTION

Password security is a core component of information security, since passwords frequently govern access to systems, applications, accounts, archives, and documents. In cybersecurity, **password cracking** refers to the process of attempting to recover a password from a protected file or its stored verification representation. Within an authorized security assessment, password cracking is used to evaluate whether passwords are sufficiently strong and resistant to common guessing techniques.

John the Ripper is an open-source password-security auditing and password-recovery tool developed by Openwall. Its Jumbo edition supports a wide range of password hashes and encrypted file formats, including password-protected PDF documents. Openwall's documentation explains that certain document formats must first be processed by a format-specific extraction utility (commonly referred to as a **2john* utility, e.g. *pdf2john*) before the resulting data can be supplied to John for candidate testing.

Networkwalks similarly presents password cracking as a hands-on cybersecurity laboratory activity designed to demonstrate how weak passwords can be recovered, and why strong passwords are necessary. Its browser-based laboratory follows the same conceptual workflow: extracting a crackable representation from a locked file and then using a password-cracking engine to identify the original password.

This Week Three project provided a practical opportunity to move from the theoretical understanding of password security into a controlled, hands-on laboratory implementation using two different toolchains — one local (John the Ripper / Johnny) and one browser-based (the Networkwalks laboratory).

3. PROJECT OBJECTIVES

The major objectives of this Week Three project were to:

- 1 Understand the concept and purpose of password cracking.
- 2 Understand the difference between an encrypted document and a crackable password representation (hash).
- 3 Install and configure John the Ripper Jumbo on a Windows workstation.
- 4 Configure the Johnny graphical interface to communicate with John the Ripper.
- 5 Prepare an encrypted PDF document for password-recovery testing.
- 6 Extract a PDF-compatible hash from a protected file.
- 7 Save and load an extracted hash into Johnny.
- 8 Perform a controlled, authorized password-recovery attack.
- 9 Understand the role of dictionaries and password-candidate lists.
- 10 Use the Networkwalks web-based password-cracking laboratory.
- 11 Recover the passwords protecting the supplied training PDF files.
- 12 Validate recovered passwords by successfully opening the protected PDFs.
- 13 Document the procedures and results as evidence of practical cybersecurity training.
- 14 Understand the defensive lessons associated with weak and predictable passwords.

4. TECHNICAL BACKGROUND

4.1 What Is Password Cracking?

Password cracking is the process of attempting to determine the password associated with a protected resource. It does not necessarily involve directly opening a protected file and repeatedly guessing through its normal interface. In most security-testing scenarios, the tester first obtains an appropriate representation of the password-verification information (a *hash*) and then performs candidate testing offline. A password candidate is tested against the target hash; if the resulting cryptographic calculation matches, the candidate is considered a successful match.

This approach is significant for security testing because offline password recovery allows a very large number of candidates to be tested without repeatedly interacting with the original application.

4.2 John the Ripper

John the Ripper is a password-security auditing and password-recovery tool. According to Openwall, it supports many password-hash and encrypted-file formats, with the Jumbo version providing additional support for formats including PDF and other encrypted documents. Certain formats require a supporting extraction utility (such as *pdf2john*) before John can process them; the extraction stage produces a representation that can subsequently be supplied to John for cracking.

4.3 Johnny Graphical Interface

Johnny is a graphical interface for John the Ripper. Rather than performing the entire exercise through a command-line terminal, Johnny provides a graphical environment through which the user can configure the John executable, open password/hash files, view detected targets, start and monitor attacks, view recovered passwords, and monitor cracking progress. During this laboratory exercise, Johnny was configured to point to the extracted John the Ripper Jumbo executable.

4.4 Networkwalks Web-Based Laboratory

The Networkwalks laboratory provides a browser-based **Hash Calculator** and **Password Cracker**. The Hash Calculator accepts a password-protected PDF, parses it locally in the browser, and extracts a pdf2john / hashcat-compatible crackable hash beginning with the \$pdf\$ marker. The extracted hash is then passed to the Password Cracker, which tests a list of password candidates against the hash and reports a match when one is found.

5. MODULE ONE — JOHN THE RIPPER & JOHNNY PASSWORD CRACKER

5.1 Module One Target

For Module One, the target file was **Locked PDF 3** ("My Locked PDF3.pdf"), supplied as part of the Networkwalks training exercise and used strictly for authorized laboratory testing. The objective was to recover the password protecting the PDF and to demonstrate the complete local password-recovery workflow using John the Ripper and the Johnny GUI.

5.2 Extracting a Crackable Hash from Locked PDF 3

The protected PDF could not be opened normally without a password. The file was therefore selected and submitted to a pdf2john-based online PDF hash extraction tool to generate a crackable representation of the PDF's password-protection information, rather than treating the protected file as a plain-text password file.

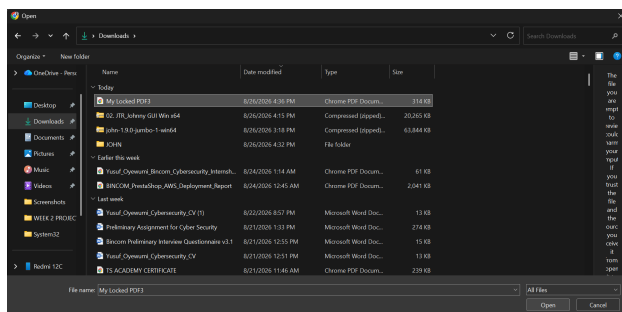


Figure 5.1 — Selecting "My Locked PDF3" for upload to the PDF hash-extraction tool.

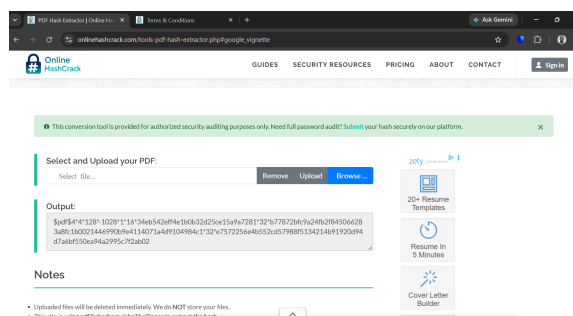


Figure 5.2 — The PDF Hash Extractor tool generates a pdf2john-compatible hash beginning with the \$pdf\$ marker, confirming that the extraction targeted PDF password-protection data.

5.3 Saving the Extracted Hash for Johnny

The extracted hash text was copied and saved as a plain-text file (HASH 1.txt) so that it could subsequently be loaded into Johnny. The saved file begins with the \$pdf\$ marker, confirming that the information was correctly recognized as a PDF password-recovery target.

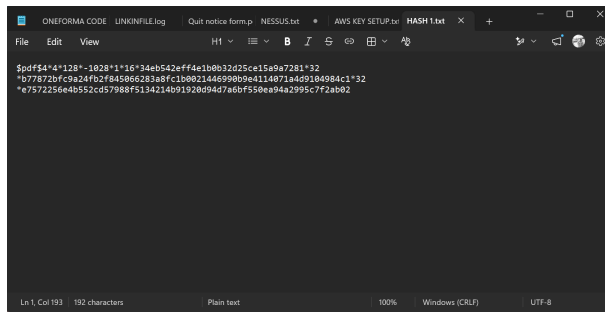


Figure 5.3 — The extracted PDF hash saved as HASH 1.txt, ready for loading into Johnny.

5.4 Installing John the Ripper and Configuring Johnny

The Windows 64-bit John the Ripper Jumbo package was downloaded and extracted into a working directory. The run directory was located and the presence of the John executable was confirmed. The Johnny graphical interface was then opened, and within Johnny's **Settings** section the path to the John the Ripper executable (`john.exe`) was browsed to and selected. Johnny successfully detected the installed John the Ripper Jumbo 1.9.0 environment, confirming that the application was ready to process password/hash files.

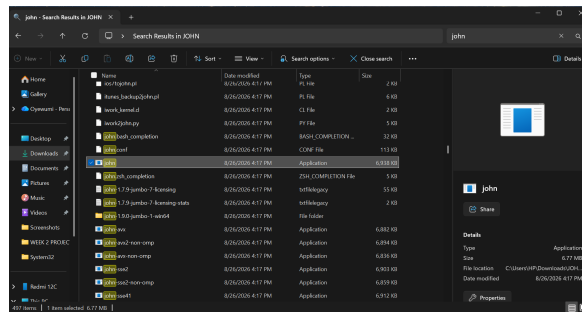


Figure 5.4 — Extracted John the Ripper Jumbo directory in Windows Explorer, with the john.exe executable located inside the run folder.

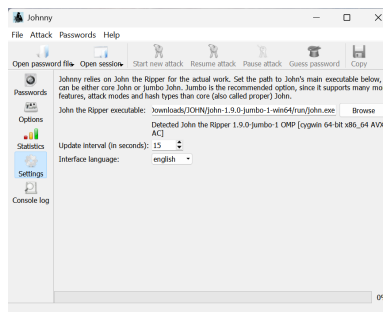


Figure 5.5 — Johnny Settings panel showing the configured path to `john.exe` and the detected "John the Ripper 1.9.0-jumbo-1" environment.

5.5 Loading the Hash into Johnny

With Johnny configured, the saved hash file was opened through Johnny's password-file interface. The file dialog was used to browse to and select the previously saved HASH_1 text file, which was then loaded into Johnny.

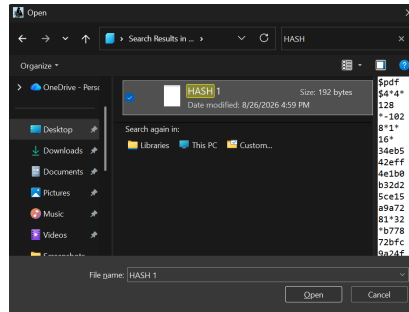


Figure 5.6 — Opening the saved HASH 1 file within Johnny's "Open password file" dialog.

5.6 Starting the Attack and Result

Once the hash had been loaded successfully, Johnny identified the target as a PDF entry and listed it in the Passwords panel. The **Start New Attack** option was then selected, and Johnny began testing password candidates against the extracted PDF hash. The attack continued until a matching candidate was identified. Johnny reported a successful crack, showing the status **1/1 cracked, 0 left** and the recovered password.

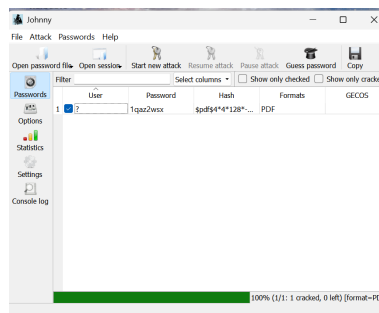


Figure 5.7 — Johnny's Passwords panel showing the target cracked, with the recovered password 1qaz2wsx and status "100% (1/1: 1 cracked, 0 left) [format=PDF]".

5.7 Validation of the Recovered Password

The recovered password was entered into the original protected PDF to confirm that it actually unlocked the document, rather than simply accepting Johnny's reported output. The PDF opened successfully, and the Networkwalks laboratory flag was captured on the unlocked page.

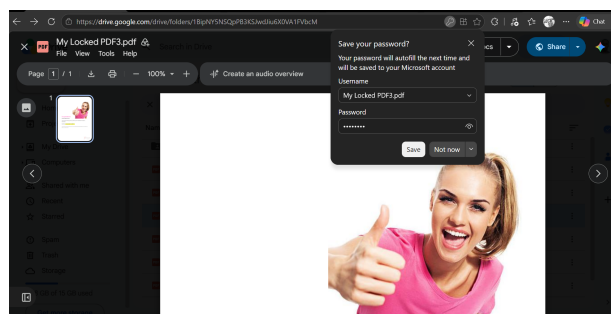


Figure 5.8 — Entering the recovered password (1qaz2wsx) to unlock "My Locked PDF3.pdf".

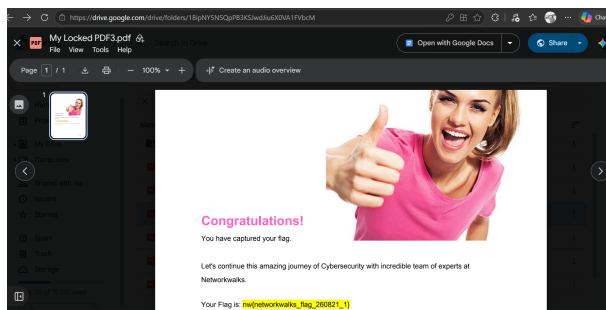


Figure 5.9 — Locked PDF 3 opened successfully, confirming recovery of the flag.

5.8 Module One Result

Target file	Locked PDF 3
Tool chain	Online pdf2john hash extractor -> saved hash file -> Johnny -> John the Ripper
Recovered password	1qaz2wsx
Status	Successfully cracked (1/1 cracked, 0 left)
Laboratory flag	nw{networkwalks_flag_260821_1}

Module One conclusion: Module One successfully demonstrated the complete local password-recovery workflow: Encrypted PDF → PDF hash extraction → saved hash file → Johnny → John the Ripper → password-candidate testing → recovered password → validation.

6. MODULE TWO — NETWORKWALKS PASSWORD CRACKER

6.1 Module Two Target

For Module Two, the target file was **Locked PDF 2** ("My Locked PDF2.pdf"). This module was performed entirely within the Networkwalks browser-based password-cracking laboratory, using the Networkwalks Hash Calculator and Password Cracker rather than a locally installed John the Ripper / Johnny workflow. The purpose of this module was to understand the same fundamental password-recovery concept through a different, web-based laboratory environment.

6.2 Hash Extraction Using the Networkwalks Hash Calculator

The Networkwalks Hash Calculator page (networkwalks.com/hash-calculator) was opened and the PDF tab was selected. Locked PDF 2 was submitted to the tool, which parsed the file locally in the browser and confirmed that the PDF was encrypted before extracting a crackable hash.

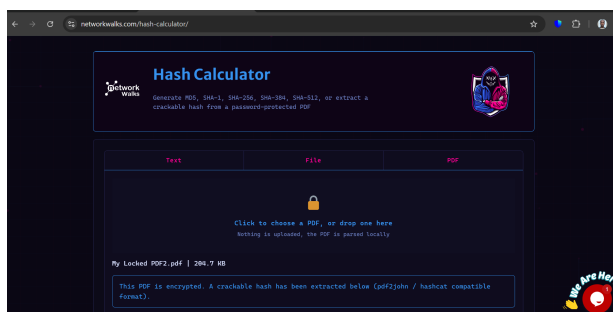


Figure 6.1 — The Networkwalks Hash Calculator (PDF tab) with "My Locked PDF2.pdf" loaded for local parsing.

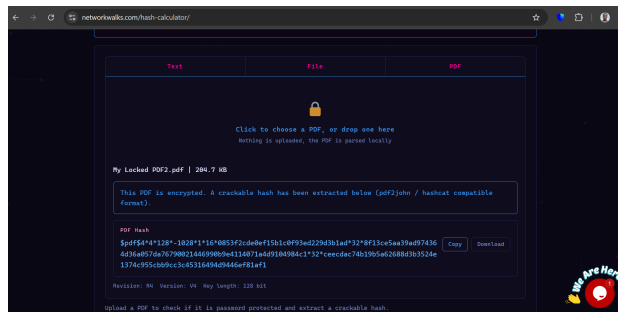


Figure 6.2 — The resulting pdf2john / hashcat-compatible PDF hash generated for Locked PDF 2, beginning with the \$pdf\$ marker (Revision: R4, Version: V4, Key length: 128 bit).

6.3 Submitting the Hash to the Password Cracker

After the hash had been generated, it was carried forward into the Networkwalks Password Cracker stage. The tool then began testing a list of password candidates against the extracted PDF hash, displaying each attempt and its outcome in real time until a matching candidate was identified.

6.4 Successful Password Recovery

The Networkwalks password-cracking process successfully identified the password after 91 of 100 candidates had been tested. The tool reported **PASSWORD CRACKED SUCCESSFULLY** with the recovered value.

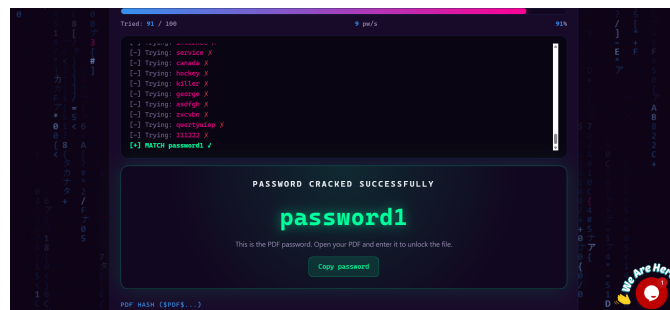


Figure 6.3 — Networkwalks Password Cracker output: candidate log showing the match "[+] MATCH password1" and the confirmed cracked password password1.

6.5 Validation of the Recovered Password

The recovered password was entered into the original protected PDF (Locked PDF 2) to confirm that it genuinely unlocked the document. This step was important because a recovered password is not considered fully verified merely because the cracking interface displays a match — opening the original protected document demonstrates that the candidate actually works against the target file.

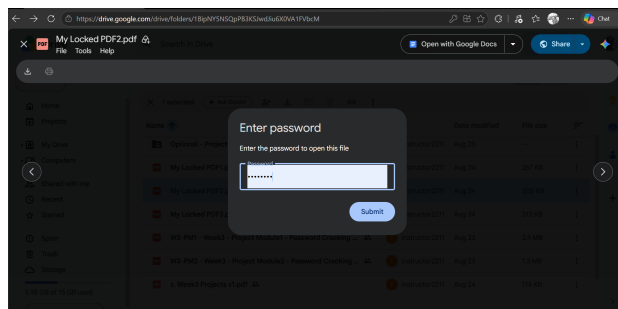


Figure 6.4 — Entering the recovered password into the "Enter password" prompt for "My Locked PDF2.pdf".

The PDF opened successfully, and the Networkwalks laboratory displayed the Module Two completion flag.

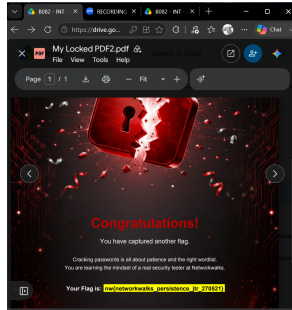


Figure 6.5 — Locked PDF 2 opened successfully, confirming recovery and displaying the completion flag.

6.6 Module Two Result

Target file	Locked PDF 2
Tool chain	Networkwalks Hash Calculator -> Networkwalks Password Cracker
Recovered password	password1
Status	Successfully recovered and validated by opening the PDF
Laboratory flag	nw{networkwalks_persistence_jtr_270521}

Module Two conclusion: Module Two demonstrated the same fundamental password-recovery workflow through a browser-based laboratory: Encrypted PDF → Networkwalks Hash Calculator → extracted hash → Networkwalks Password Cracker → recovered password → validation.

7. COMPARISON OF MODULE ONE AND MODULE TWO

Category	Module One	Module Two
Target	Locked PDF 3	Locked PDF 2
Environment	Windows workstation (local)	Web browser
Main tool	John the Ripper + Johnny	Networkwalks Hash Calculator + Password Cracker
Hash preparation	Online pdf2john hash extractor	Networkwalks Hash Calculator (client-side parsing)
Attack	Local password-recovery attack	Web-based laboratory recovery
Recovered password	1qaz2wsx	password1
Validation	Johnny showed 1/1 cracked; PDF opened	PDF opened successfully
Flag	nw{networkwalks_flag_260821_1}	nw{networkwalks_persistence_jtr_270521}

The two modules used different interfaces but demonstrated the same fundamental security concept: obtain an appropriate crackable representation of the protected resource, test password candidates against it, recover a matching candidate, and validate the result against the original file.

8. OBSERVATIONS AND FINDINGS

8.1 Weak Passwords Can Be Recovered Quickly

The passwords recovered during the laboratory exercise were simple and predictable: **1qaz2wsx** (Module One) is a common keyboard-pattern password, and **password1** (Module Two) is one of the most frequently used passwords worldwide. Both examples demonstrate why predictable passwords present a significant security risk.

8.2 Password Length and Complexity Matter

A password that is short, common, predictable, or based on a keyboard pattern can be significantly easier to recover than a long and unique passphrase. The exercise reinforced the importance of using passwords that are long, unique, difficult to predict, not based on common words, not reused across systems, and generated or stored using appropriate password-management practices.

8.3 Offline Password Attacks Are Important

One of the most important lessons from the exercise was the difference between online and offline password attacks. Once an appropriate password-verification representation has been obtained, candidate testing can occur entirely offline, meaning an attacker does not need to repeatedly interact with the original PDF or application. For defenders, this highlights the importance of protecting password hashes, encrypted files, backups, and other authentication-related data.

8.4 Correct Tool Configuration Is Important

Successful password recovery depended on proper preparation at every stage: John installation → Johnny configuration → PDF hash extraction → hash-file creation → hash loading → attack configuration → attack execution → validation. A failure at any single stage could have prevented the password-recovery process from working correctly.

8.5 Result Validation Is Necessary

In Module One, Johnny reported a successful crack (1/1 cracked). In Module Two, the recovered password was entered into the protected PDF and the document opened successfully. In both cases, opening the original protected file provided stronger evidence of success than simply accepting a tool's reported output at face value.

9. SKILLS ACQUIRED DURING THE PROJECT

- Password security concepts and terminology.
- Password-cracking methodology and workflow.
- Practical use of John the Ripper (Jumbo edition).
- Use of the Johnny graphical interface.
- PDF password-hash extraction (pdf2john-based tools).
- Hash-file preparation and formatting.
- Dictionary / candidate-based password recovery.
- Web-based password auditing using Networkwalks tools.
- Basic technical troubleshooting during tool configuration.
- Evidence collection and documentation.
- Result validation methodology.
- Ethical hacking laboratory practices.
- Cybersecurity reporting and write-up skills.

- Understanding the relationship between weak passwords and security exposure.

10. CHALLENGES ENCOUNTERED

10.1 John / Johnny Configuration

Johnny needed to be correctly configured to locate the John the Ripper executable. Selecting the wrong executable or directory could prevent the graphical interface from communicating correctly with John.

10.2 Correct Hash Format

The extracted PDF information had to be saved in a format that John could recognize. The presence of the \$pdf\$ marker in the resulting data helped confirm that the extraction process had produced PDF-specific cracking information.

10.3 Correct File Selection

The project contained multiple locked PDF files. Care was required to ensure that the appropriate file was used for each module: **Module One = Locked PDF 3, Module Two = Locked PDF 2.**

10.4 Verification

After each password was recovered, the result had to be validated against the original protected document. This prevented an incorrect candidate or a misinterpreted tool output from being recorded as a successful result.

11. DEFENSIVE SECURITY RECOMMENDATIONS

11.1 Use Strong Passwords

Organizations and individuals should use long, unique, unpredictable passwords rather than short or common passwords.

11.2 Avoid Keyboard Patterns

Patterns such as 1qaz2wsx illustrate why keyboard-based sequences should not be used for sensitive files or accounts.

11.3 Avoid Common Passwords

Passwords such as password1 are highly predictable and should never be used to protect sensitive information.

11.4 Use Password Managers

Password managers can generate and securely store unique passwords, reducing the temptation to reuse simple passwords.

11.5 Use Multi-Factor Authentication

Where supported, MFA should be enabled for accounts that provide access to sensitive systems or data.

11.6 Protect Extracted Hashes

Hash files created during security assessments should be treated as sensitive information and stored securely.

11.7 Delete Temporary Assessment Data

After an authorized password audit, temporary hash files, recovered passwords, and other assessment artifacts should be securely removed when no longer required.

11.8 Conduct Authorized Password Audits

Organizations should periodically perform authorized password-strength assessments to identify weak credentials before attackers do.

12. ETHICAL AND LEGAL CONSIDERATIONS

Password-cracking tools are dual-use cybersecurity technologies. They can be used by security professionals to test password strength, recover authorized passwords, audit organizational security, conduct penetration tests, demonstrate cybersecurity vulnerabilities, and improve defensive controls. However, the same techniques can be misused if applied to accounts, documents, networks, or systems without authorization.

For this reason, all password-recovery activities in this project were limited strictly to the training files supplied as part of the Networkwalks internship laboratory. A professional cybersecurity practitioner should always ensure that:

- 1 Authorization is obtained before testing.
- 2 The scope of testing is clearly defined.
- 3 Test data is handled securely.
- 4 Recovered credentials are not unnecessarily disclosed.
- 5 Temporary assessment artifacts are protected.
- 6 Testing is stopped once the authorized objective has been achieved.

Networkwalks presents its password-cracking activities within a cybersecurity training context, while Openwall describes John the Ripper as a password-security auditing and password-recovery tool.

13. OVERALL PROJECT RESULTS

Module One

Target	Locked PDF 3
Tool	John the Ripper + Johnnny
Recovered Password	1qaz2wsx
Status	Successfully cracked
Laboratory Flag	nw{networkwalks_flag_260821_1}

Module Two

Target	Locked PDF 2
Tool	Networkwalks Hash Calculator + Password Cracker
Recovered Password	password1
Status	Successfully recovered and validated by opening the PDF
Laboratory Flag	nw{networkwalks_persistence_jtr_270521}

The Week Three password-cracking project was successfully completed, with both modules yielding correctly recovered and validated passwords, and both laboratory flags captured.

14. CONCLUSION

Week Three of my Networkwalks cybersecurity internship provided valuable practical experience in password security, password recovery, and ethical hacking. The project began with a theoretical understanding of password cracking and progressed into practical implementation using two different approaches.

In **Module One**, I installed and configured John the Ripper and Johnny on a Windows workstation. I learned how to prepare an encrypted PDF, extract its PDF-specific password-recovery information, save the resulting hash, load the hash into Johnny, start a controlled password-recovery attack, and validate the recovered password. In **Module Two**, I applied the same fundamental concept through the Networkwalks web-based password-cracking laboratory, extracting the hash from Locked PDF 2, submitting it to the Password Cracker, recovering the password, and verifying the result by opening the protected document.

The project demonstrated that password security is not simply about whether a document has a password: the **quality, length, uniqueness, and unpredictability** of the password are critical factors in determining how resistant it is to password-recovery attacks. The successful recovery of **1qaz2wsx** and **password1** illustrated the weakness of predictable passwords and provided a practical lesson in why stronger authentication controls are required.

Overall, the project strengthened my technical understanding of the full workflow: PDF encryption → hash extraction → password candidates → password recovery → validation → security analysis, and has provided a solid foundation for more advanced work in ethical hacking, penetration testing, password auditing, vulnerability assessment, and defensive security.

15. REFERENCES

- 1 Openwall — John the Ripper official project: <https://www.openwall.com/john/>
- 2 Openwall — John the Ripper Documentation: <https://www.openwall.com/john/doc/>
- 3 Openwall — John the Ripper Installation Documentation:
<https://www.openwall.com/john/doc/INSTALL.shtml>
- 4 Openwall — John the Ripper FAQ: <https://www.openwall.com/john/doc/FAQ.shtml>
- 5 Networkwalks Academy — Project Task Lab: Password Cracking with Networkwalks Tools:
<https://networkwalks.com/project-task-lab-password-cracking-with-networkwalks-tools/>
- 6 Networkwalks Academy — Official Website: <https://networkwalks.com/>

APPENDIX A — PRACTICAL EVIDENCE INDEX

The figures embedded throughout this report constitute the practical evidence supplied during the Week Three project. They are indexed below for reference.

Figure	Description
5.1	Locked PDF 3 selected for hash extraction
5.2	PDF Hash Extractor output (\$pdf\$ hash) for Locked PDF 3
5.3	Extracted hash saved as HASH 1.txt

5.4	Extracted John the Ripper Jumbo files (Windows Explorer)
5.5	Johnny Settings — John the Ripper executable configured
5.6	HASH 1 file opened within Johnny
5.7	Johnny attack result — password 1qaz2wsx cracked (1/1)
5.8	Recovered password entered into Locked PDF 3
5.9	Locked PDF 3 opened — Module One flag captured
6.1	Networkwalks Hash Calculator — Locked PDF 2 loaded
6.2	Networkwalks Hash Calculator — extracted PDF hash for Locked PDF 2
6.3	Networkwalks Password Cracker — password1 cracked
6.4	Recovered password entered into Locked PDF 2
6.5	Locked PDF 2 opened — Module Two flag captured

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